HASSAM ASLAM

Gameplay & AI Programmer · Unreal Engine 5 (C++ / Blueprints)

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SUMMARY

Gameplay programmer working in UE 5.8 C++ and Blueprints, focused on combat, the Gameplay Ability System, enemy and NPC AI, and multiplayer. Shipped 12 Fortnite islands as Zeritress (855M+ minutes played). I build systems designers can tune through data, and I prove they work with automation tests and measured results.

SKILLS

Engine & code	Unreal Engine 5 (to 5.8), C++, Blueprints, Verse (UEFN), Visual Studio, Git
Gameplay	Gameplay Ability System (abilities, effects, attribute sets, tags, cues, prediction), combat, input buffering, cameras, Enhanced Input
AI	AI Controllers, behavior trees with custom C++ nodes, EQS, AI Perception, NavMesh and Detour crowd, LLM integration and retrieval
Multiplayer	Server-authoritative replication, client prediction, lag compensation, fast-array replication, listen and dedicated servers
UI, tools & testing	UMG written in C++, editor commandlets that generate assets, automation tests, debug overlays and console commands

EXPERIENCE

UEFN Developer · Zeritress, freelance with creator partners — 2023 – present

- Shipped **12 Fortnite islands** with 6 creator partners: **855M+ minutes played**, a 44,962 all-time player peak, 482 published versions.
- Built Verse systems: persistent economies and save state, shops and item granting, per-player state tested with 50 players, prestige and ranked progression.
- Ran live-ops at about two updates a week on flagship islands, including PlayStation vs Xbox (552M minutes played).

AI Programmer & Technical Animator · Moshpit Studios

- NPC behaviour for a multiplayer virtual concert with AI Controllers and behavior trees; technical animation for crowd and player characters; tested NPCs against player powers.

C++ Contributor · Malakhra: Blood & Mayhem, 6-person team, UE 5.8 — 2026

- Built menu-hub systems in C++ (fighter cages, vanity wall, options, clickable navigation) and an editor-only camera-anchor visualizer for level artists.

SELECTED PROJECTS

Field Correspondent · solo, UE 5.8 C++ — 2026

- LLM-driven NPC: the model proposes actions through tool calls, C++ checks them against a whitelist and the NavMesh, and a behavior tree with six custom C++ nodes carries them out. Interrupts in about 0.2 s.
- Hybrid retrieval with citations (0.044 ms median search), AI Perception into timed memories. **120 automation tests** and 50 end-to-end checks passing in a packaged build.

Ashford Hollow · solo, UE 5.8 C++, 2–4 player networked RPG — 2026

- About 18,600 lines of C++: GAS combat (melee combo with input buffering, predicted ranged attack, area slam, dash strike), server-authoritative hits and loot, a quest-giver NPC.
- Enemy AI: one AI Controller with Perception, a behavior tree per enemy type, EQS positioning, and attack tokens for a pack that flanks.
- Server-side lag compensation that **cut ghost hits from 30% to 2% at 300 ms**, measured with a scripted test client.

GASShooter port to UE 5.8 — 2026

- Ported tranek's GAS shooter sample from 5.3 to 5.8 (removed montage replication fields, instancing policy changes, prediction keys) and fixed an inverted ensure that blocked server-side target-data callbacks.

Third-person action RPG · final year project, UE5

- Melee combat with elemental abilities, enemy AI, XP and a three-branch skill tree across two levels.